AI-assisted Software Development

Trends and Lessons

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GitHub Next

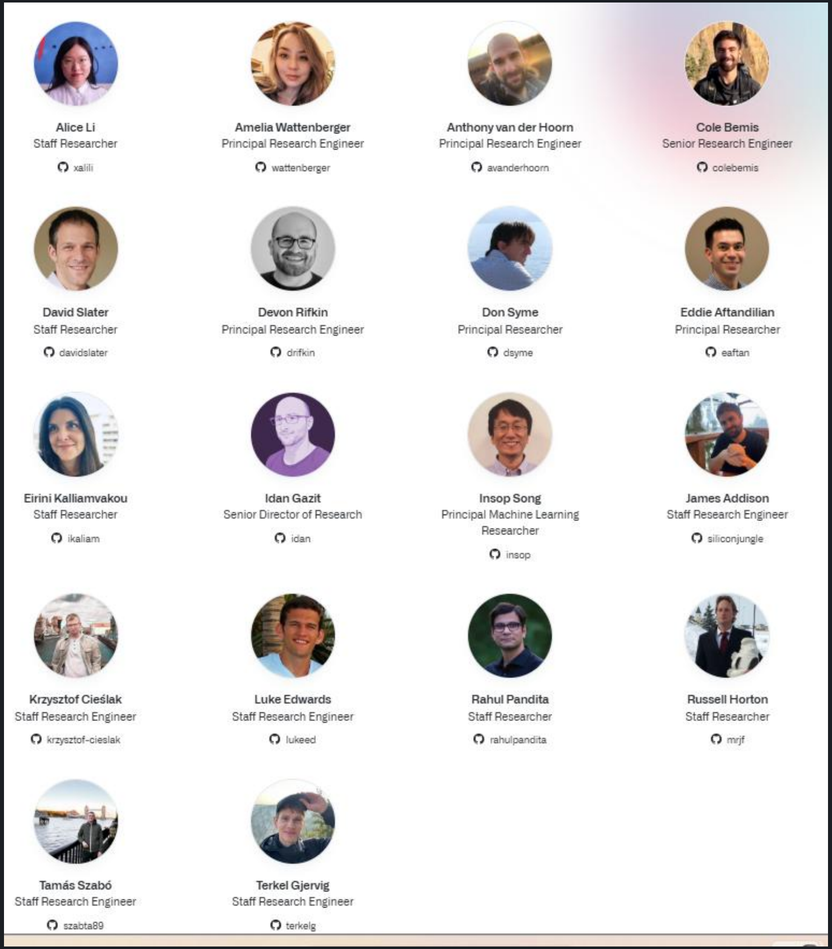
Research, Build, Ship, Learn

GitHub Next

GitHub Next investigates the future of software development.

We are a team of researchers and engineers at GitHub, exploring things beyond the adjacent possible. We prototype tools and technologies that will change our craft. We identify new approaches to building healthy, productive software engineering teams.

The GitHub Next Team



Your Big Questions!

“What will coding with AI look like in the future?”

“Does AI mean that code will be obsolete soon?”

“How should I use AI tools in software development?”

We live in the most incredible of times

- **AI**, **indexing** and **traditional dev tools** make us all magicians when it comes to code

- But the magic power comes with many difficulties!

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Some Magic Wands

cody

Meet Cody Your
AI-Assistant

CURSOR

GitHub
Copilot

AI code completions

JS test.js 1

JS test.js > calculateDaysBetweenDates

```
function calculateDaysBetweenDates(begin, end) {
    var beginDate = new Date(begin);
    var endDate = new Date(end);
    var days = Math.round((endDate - beginDate) / (1000 * 60 * 60 * 24));
    return days;
}
```

AI Dev Tools are changing fast

- 2022: **GitHub Copilot** launched. Today:
  - 1.8M+ paying users
  - 50%+ of users' code is written by Copilot
- 2023: More AI developer tools launched
  - Summarization, advice, code review
  - Cody and others
- 2024: **GitHub Copilot Workspace, Cursor Composer, Devin** launched
  - Shift to task-orientation
  - Cursor Composer brought task-orientation to the IDE
- 2025: Cursor/VS Code **Agent-Mode**, **Claude Code** and others launched
  - Increases the available automation
- 2025: **Automated app development** starts to take off
  - GitHub Spark, Bolt, Gemini, Lovable, Cogna, ...

Where does GitHub Next work?

AI APPLICATIONS & AI DEV TOOLS

AI ENRICHED SERVICES (e.g. indexing, summaries)

FINE TUNING & INFERENCE & GPUS & HOSTING & ....

MODELS

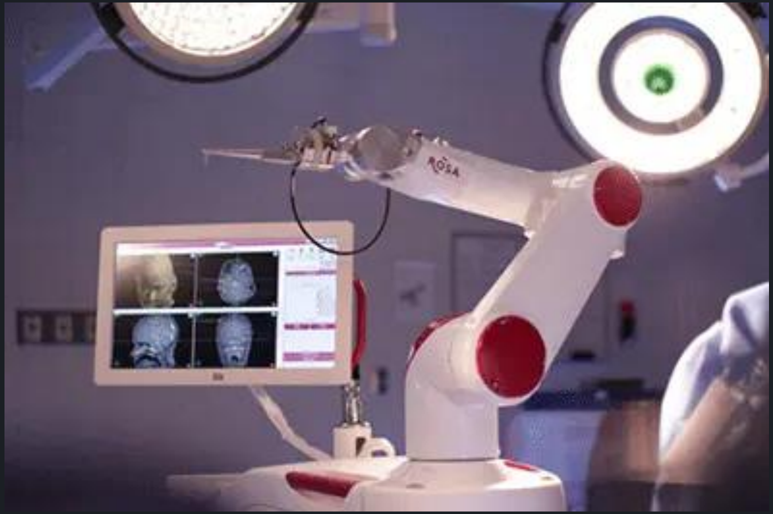
The Big Trends I'm seeing in AI Dev Tooling

- AI-assisted Pinpoint Surgery
- AI-assisted Task Oriented Surgery
- Increasing AI Automation
- AI-Assisted App Development

Theme 1: Pinpoint Surgery

Original GitHub Copilot completes code in your editor

- You are already located at a pinpoint of change
- You accept/reject small bits
- You have to type quite a lot
- Copilot sort of infers what you are trying to do



This is magical: a finely honed surgical assistant

Other Pinpoint Tools

- Type-checking
- Autocomplete
- On-the-fly checking
- Code navigation
- Code search
- Security checking

Other Pinpoint Tools

- Type-checking
- Autocomplete
- On-the-fly checking
- Code navigation
- Code search
- Security checking
- **AI Code completion**
- **AI Code summarization**
- **AI code chat**
- **AI Code review**
- **AI Code-fix suggestions**
- **AI Test generation**

Theme 2: From Pinpoint Surgery to Tasks

Software activity begins with words and intent

Before a code change, before a pull request, there are words - a discussion, a bug report, a feature request, an engineering task, a meeting, a design document, an idea.

Theme 2: From Pinpoint Surgery to Tasks

https://www.flaticon.com/free-icons/

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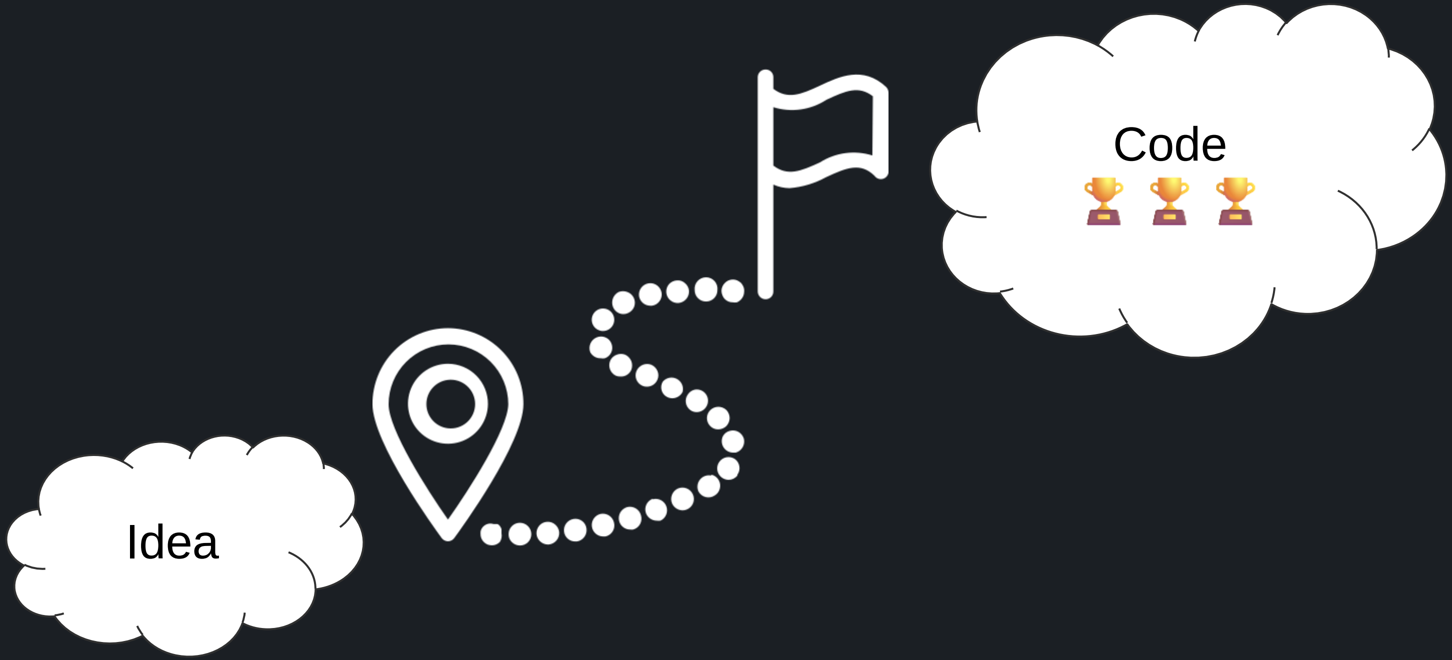

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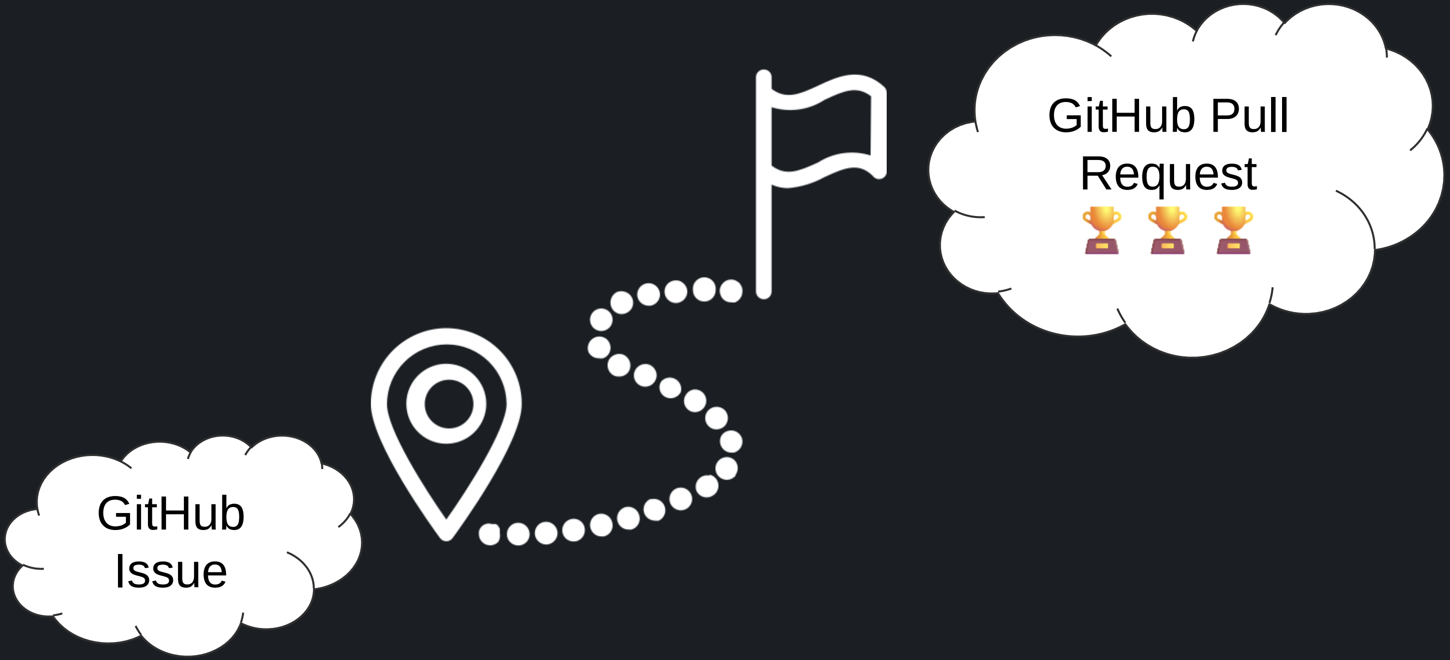

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Theme 2: From Pinpoint Surgery to Tasks

Changing perspective

Pinpoint Surgery → Whole Operation

Single Point of Change → Whole Repository

Implicit Task →

Theme 2: From Pinpoint Surgery to Tasks

“Add a refresh button to the main page”

“Create end-to-end tests using Playwright”

“Set up continuous integration using GitHub Actions”

“Set up production resources in Azure using Terraform”

“Read coverage report and add unit tests to repo”

“Create a website for UK house prices”

Demo - Repo Tasks

Theme 3: Automation with SWE-Agents

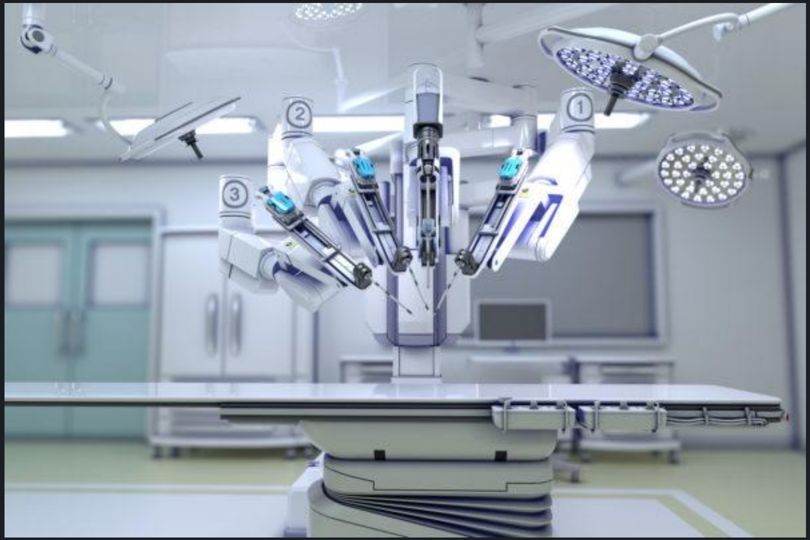
SWE-Agents are AI agents that have multiple tools to operate on your code.

Can be unreliable, the “sweet zone” requires skill to find.

But improving all the time and competent at some things.

Learn and use them!

Don’t let them operate on you!



Theme 4: AI-Assisted App Development

"Create me a multiplayer online TicTacToe game"

Gemini

bolt

GitHub Spark

Cogna

lovable

Theme 4: AI-Assisted App Development

- Narrows the scope to **app development**
- Usually narrows to particular **design theme** and **backend services**
- The user can **run the app** and **validate directly**
- Sometimes **little need to look at the code at all** (vibe coding)

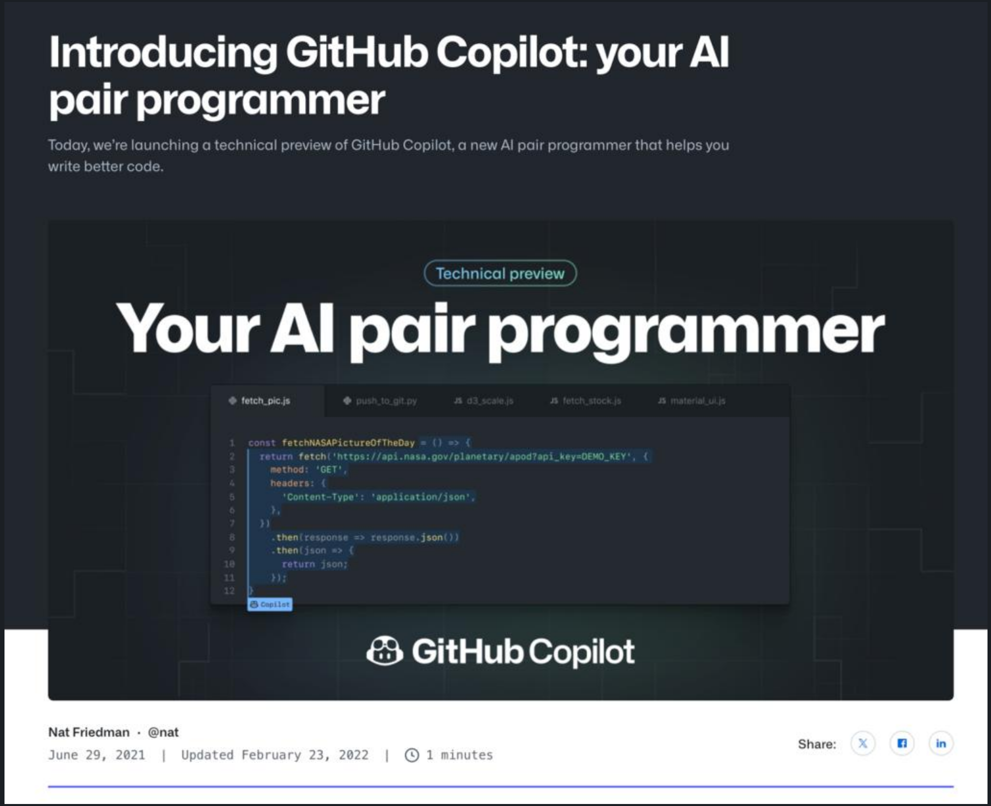
Huge caveats

- **App coding** is replaced by **app description**
- This is a small but important slice of the software industry
- Good **UX design** is **subjective**, impossible to describe with words
- Beware security disasters!

Demo - GitHub Spark

General lessons

Lesson: A small team can build a big thing



Lesson: Build a prototype ASAP

Lesson: Deliver tangible results frequently

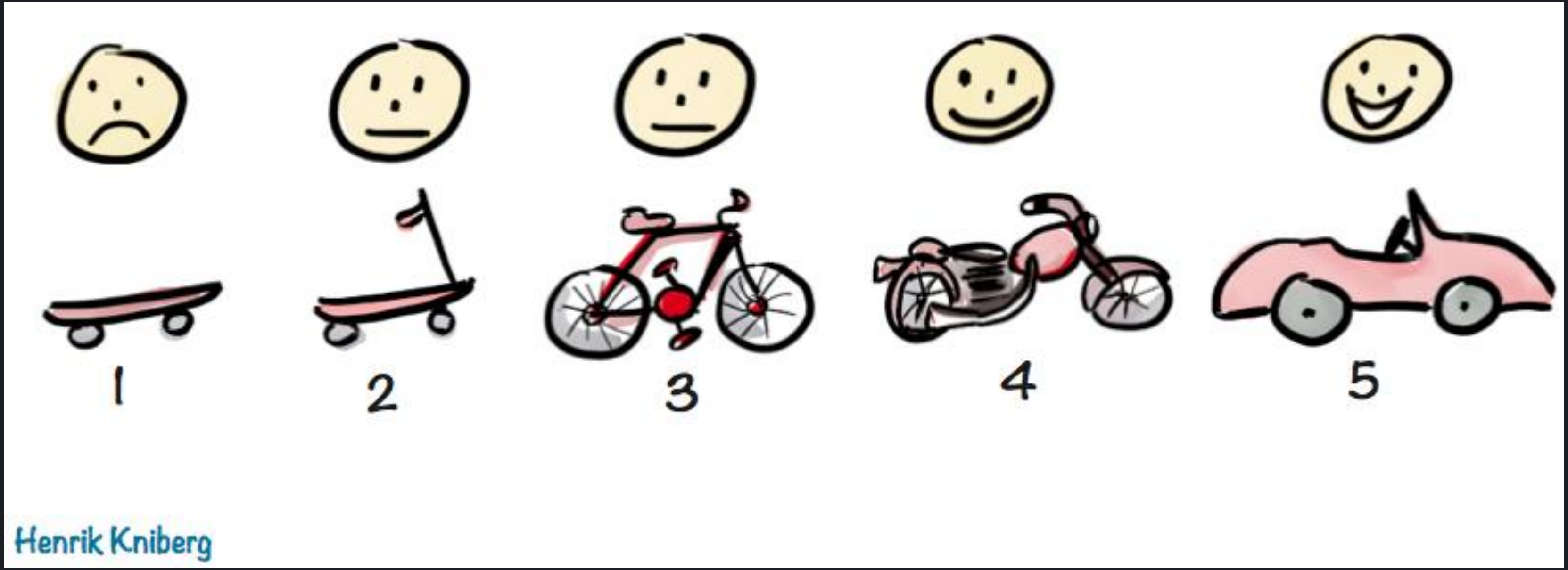
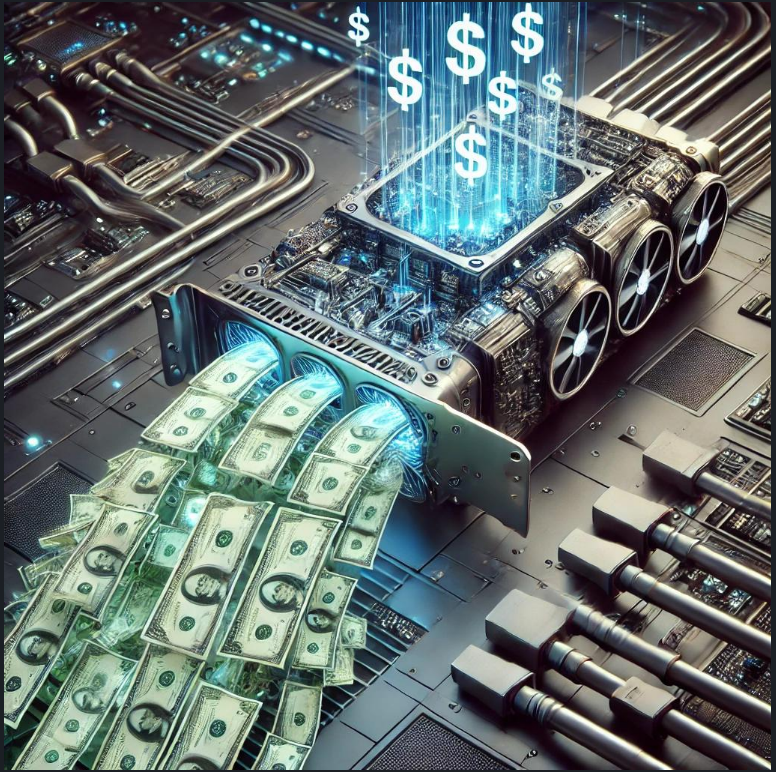


Image credit: https://blog.crisp.se/2016/01/25/henrikkniberg/making-sense-of-mvp

Lesson: UX is critical in ~~AI~~ developer tools

Lesson: Beware of long term product-market fit

Lesson: Don’t over index on cost



Lesson: Dogfood intensively

Your Questions, My Answers

My Answers!

“What will coding with AI look like in the future?”

1. AI-powered **pinpoint tools** will be everywhere
2. AI-assisted **task orientation** will become normal
3. Some **technical coding** will be fully automated and become **technical description + exploration**
4. UX design is **subjective** and will be only partially automated
5. **Requirements**, **validation**, **architecture**, **maintainability**, **improvement**, **security** will be only partially automated

My Answers!

"Does AI mean that code will be obsolete soon?"

1. No! Code will largely remain the record of ground truth
2. **Technical description** will become more important
3. **Depth knowledge** will continue to be critical
4. **Requirements** and **validation** will begin to dominate

My Answers!

"How should I use AI tools in software development?"

1. Learn to be a great programmer **without AI**
2. **Embrace** many pinpoint tools
3. **Learn** task-oriented tools
4. **Play with** automation like Claude Code
5. **Play with** app-dev tools like Lovable and GitHub Spark

We live in the most incredible of times

- Coding is already magical, AI coding gives you wild powers
- These powers are not simple to use well
- AI is changing coding fast, and good developers embrace some of it

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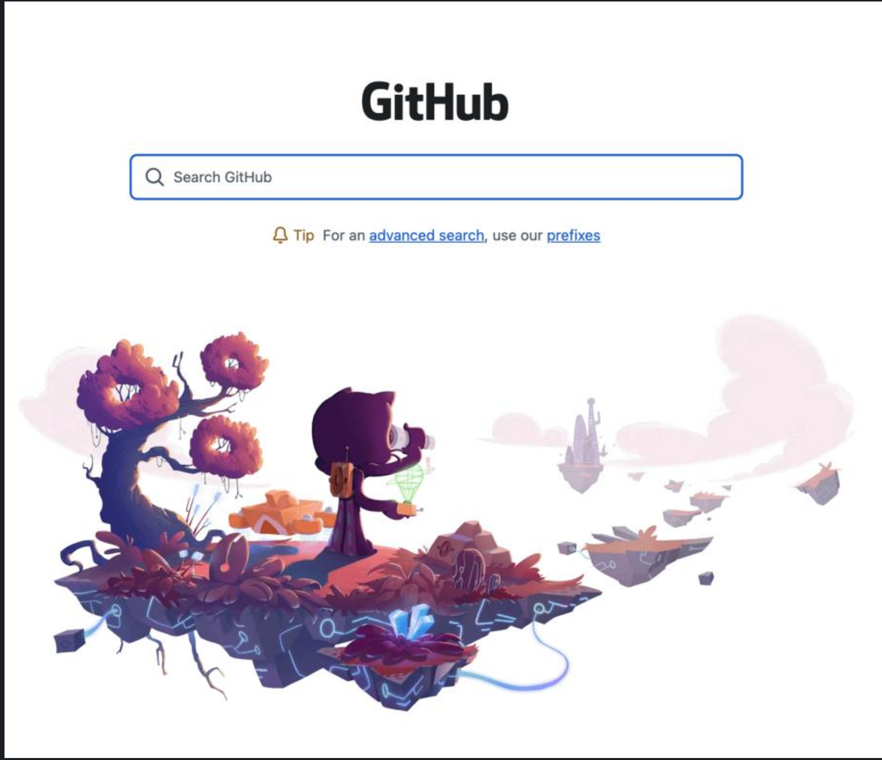
GitHub Next

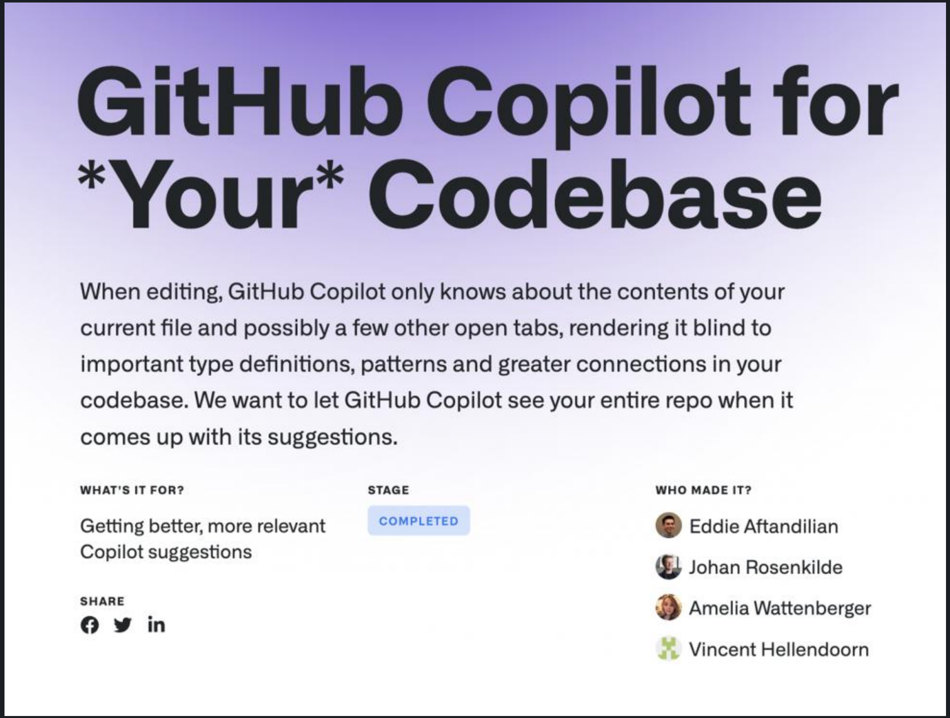
Investigating the future of software development

githubnext.com

Retrieval application 1: NL code search



Retrieval application #2: customized Copilot



Retrieval application #3: Copilot for Docs